

$$* \sum_{n \geq 0} U_n \text{ où } U_n = \sin(\pi(2+\sqrt{3})^n)$$

$$\forall n \in \mathbb{N}, (2+\sqrt{3})^n + (2-\sqrt{3})^n = \sum_{k=0}^n C_n^k 2^{n-k} (\sqrt{3})^k + \sum_{k=0}^n C_n^k 2^{n-k} (-1)^k (\sqrt{3})^k$$

$$= \sum_{0 \leq k \leq n/2} C_n^{2k} 2^{n-2k} 3^k + \sum_{0 \leq k \leq n/2} C_n^{2k+1} 2^{n-2k-1} 3^k \sqrt{3}$$

$$+ \sum_{0 \leq k \leq n/2} C_n^{2k} 2^{n-2k} 3^k + \sum_{0 \leq k \leq (n-1)/2} C_n^{2k+1} 2^{n-2k-1} 3^k \sqrt{3}$$

$$= 2A_n \text{ où } A_n = \sum_{0 \leq k \leq n/2} C_n^{2k} 2^{n-2k} 3^k \in \mathbb{N}$$

$$\text{donc } U_n = \sin(2A_n - (2-\sqrt{3})^n) = -\sin((2-\sqrt{3})^n) \sim -\pi(2-\sqrt{3})^n$$

Car $-1 \in \mathbb{R}^*$ et $0 \leq 2-\sqrt{3} < 1$ alors d'après la règle « x^n », $\sum_{n \geq 0} U_n$ converge.

$$* \sum_{n \geq 0} U_n \text{ où } U_n = \sin(\pi \underbrace{n! e^{-2}}_{w_n})$$

$$\forall n \in \mathbb{N}, w_n = \sum_{k=0}^n \frac{(-1)^k n!}{k!} + \underbrace{\sum_{k=n+2}^{\infty} \frac{(-1)^k n!}{k!}}_{w_n} \text{ car } e^{-1} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!}$$

$$\text{on a: } |w_n| \leq \sum_{k=n+2}^{\infty} \frac{1}{A_k} = \frac{1}{n+1} + \sum_{k=n+2}^{\infty} \frac{1}{A_k} \\ \leq \frac{1}{n+1} + \sum_{k=n+2}^{\infty} \frac{1}{k(k-1)} = \frac{1}{n+1} + \lim_{m \rightarrow \infty} \sum_{k=n+2}^m \left(\frac{1}{k-1} - \frac{1}{k} \right) \\ \leq \frac{1}{n+1} + \frac{1}{n+1} - 0 = \frac{2}{n+1}$$

$$\text{Ainsi } \lim_{n \rightarrow \infty} w_n = 0$$

$$V_n = \underbrace{\sum_{k=0}^n (-1)^k A_n^{n-k}}_{A_n \in \mathbb{R}} \left(1 + \omega n x \underbrace{\frac{1}{n! \sum_{k=0}^n \frac{(-1)^k n!}{k!}}}_{d_n} \right)$$

or $\lim_{n \rightarrow \infty} d_n = 0 \times \frac{1}{\infty \times e^1} = 0$ car $V_n \sim A_n = \sum_{k=0}^n (-1)^k A_n^{n-k}$

Plus encore, $U_n = \sin(\pi A_n + \pi \omega n)$

or $A_n \equiv 1 \pmod{2}$ pour n pair car dans ce cas, $A_n^{n-k} \equiv 1 \pmod{2}$

et $A_n^k \equiv 0 \pmod{2} \forall k \in \{0, n\}$

Donc n impair $A_n \equiv \sum_{k=0}^{n-2} (-1)^k A_n^{n-k} + (-1)^{n-1} x n + (-1)^n x 1 \equiv 0 \pmod{2}$

Ainsi, pour n pair, $U_n = -\sin(\pi \omega n)$

pour n impair, $U_n = +\sin(\pi \omega n)$

Donc $\forall n \in \mathbb{N}$, $U_n = (-1)^{n+2} \sin(\pi \omega n) \sim \underbrace{(-1)^{n+2}}_{B_n} \times \pi \sum_{k=n+2}^{\infty} \frac{(-1)^k n!}{k!}$

or ~~pro~~

$$\frac{1}{t^2}$$